Chapter 2: Variables and Data Types

Python Programming Notes | Beginner Study Chapter

Python variables

A variable is a name that refers to a value. Create a variable using assignment, for example age = 20. Python infers the type from the assigned value, so there is no need to declare the type. A variable may later refer to a different value. Variable names are case-sensitive and cannot begin with a digit.

Basic types

int stores whole numbers such as 20. float stores decimal numbers such as 3.5. str stores text such as "Ali". bool stores True or False. type() tells you the type of a value. int(), float() and str() convert compatible values to another type.

Lists and tuples

A list is an ordered, mutable collection created with square brackets, such as marks = [80, 90, 75]. A tuple is an ordered, immutable collection created with parentheses, such as point = (3, 4). Both support indexing from zero and allow duplicate values. You can change list items or append new ones. You cannot assign a new value to a tuple item. Use lists for changing data and tuples for fixed groups of values.

Other collections

A dictionary stores key-value pairs, for example student = {"name": "Ali", "age": 20}. A set stores unique elements. Use len() to count elements in a collection.

Try this code

```
name = "Sara"
age = 20
marks = [80, 90, 75]
marks.append(85)
point = (3, 4)
print(name, age, marks, point)
```